

Short Questions 9

04/12/2026

Your answers should be formal with mathematical formulas and algorithms, as necessary. Upload a PDF file. If you want to add a drawing, you can include a nicely hand-drawn figure or a figure drawn with a computer tool. Use single spaced typing. Preferably, keep your answers to a page.

Type your responses.

1. (3 points) How does the Decision Transformer work?
2. (3 points) What is Advantage for an action in a state? Some researchers believe using Advantage $A(s, a)$ provides better reinforcement learning than using the straightforward action value function $Q(s, a)$. What motivates the use of the advantage function?
3. (4 points) Briefly discuss two variants of Deep Q Learning, beyond Double DDN and Dueling DDN.